Sequence-Based Drug-Target Affinity Prediction: A Study of Generalisation to Unseen Drugs and Targets

Project Report submitted in partial fulfillment of the requirements for the award of the degree of

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in

Computer Science (Artificial Intelligence & Data Science)

by

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CERTIFICATE

This is to certify that the project report for ***23-813-0607: Project & Viva Voce*** *on the topic* ***Sequence-Based Drug-Target Affinity Prediction: A Study of Generalisation to Unseen Drugs and Targets*** *is a record of work carried out by* ***RISHAN P N (81323020)****, in partial fulfillment of the requirements for the award of degree in* ***Master of Science (Five Year Integrated)*** *in* ***Computer Science (Artificial Intelligence & Data Science)*** *of Cochin University of Science and Technology (CUSAT), Kochi. The project report has been approved as it satisfies the academic requirements in respect of the project prescribed for the Master of Science (Five Year Integrated) in Computer Science degree.*

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DECLARATION

I hereby declare that the project report entitled **"Sequence-Based Drug-Target Affinity Prediction: A Study of Generalisation to Unseen Drugs and Targets"** submitted as part of the curriculum for the Master of Science (Five Year Integrated) in Computer Science (Artificial Intelligence & Data Science) at Cochin University of Science and Technology is my original work. This written submission represents ideas in my own words. Where others' ideas and words have been included, I have adequately cited and referenced the original source. I declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the source which has not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Accurate prediction of drug-target affinity (DTA) is a critical step in computational drug discovery, enabling the rapid identification of promising drug candidates without the cost and time demands of experimental assays. This project proposes and implements a sequence-based deep neural network model for DTA prediction, evaluated on two widely used benchmark datasets: Davis and KIBA.

The proposed model (**ImprovedDTA**) encodes drug molecules from SMILES strings and protein targets from amino acid sequences using a shared character-level embedding layer. Drug features are extracted using a **multi-scale convolutional neural network** with parallel kernel sizes of 3, 5, and 7, while protein features are extracted through a **convolutional and bidirectional LSTM** encoder. A **multi-head cross-attention** mechanism then models the drug-target interaction by attending protein representations over drug features. The final binding affinity is predicted by a fully connected regression head with Batch Normalisation and Dropout.

The model is evaluated under two experimental protocols: (i) a **random split**, simulating a standard in-distribution setting, and (ii) an **unseen split**, testing generalisation to novel drugs, novel targets, and both simultaneously — the most realistic scenario for drug discovery. On the random split, the model achieves an MSE of **0.2489** and CI of **0.8875** on Davis, and an MSE of **0.2248** and CI of **0.8545** on KIBA. Under the more challenging unseen evaluation, the model maintains competitive concordance indices while offering dramatically simpler architecture and lower computational cost compared to state-of-the-art dual-modality graph-based models. Results are compared directly against the DMFF-DTA benchmark and its published SOTA comparison tables.

Keywords: Drug-Target Affinity, DTA Prediction, Sequence-Based Model, CNN, BiLSTM, Cross-Attention, Davis Dataset, KIBA Dataset, Unseen Generalisation.

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Chapter 1

Introduction

1.1 Background

The discovery and development of new drugs is one of the most resource-intensive endeavors in modern science. Bringing a single drug from initial target identification to market approval routinely takes 10–15 years and costs between 400 and 800 million USD [11]. A central bottleneck in this pipeline is the experimental measurement of binding affinities between drug candidates and their protein targets. Traditional biochemical assays are low-throughput, expensive, and inapplicable to the vast chemical space of novel compounds. This has motivated decades of research into computational methods capable of predicting drug-target affinity (DTA) rapidly and accurately.

Drug-target affinity quantifies the strength of the physical interaction between a small drug molecule and a target protein, typically measured as the equilibrium dissociation constant (K_d) or the half-maximal inhibitory concentration (IC_{50}). Lower values indicate stronger binding. In practice, the negative logarithm of these quantities (pKd or pIC50) is used as the continuous affinity label, making DTA prediction a regression problem. Accurate DTA prediction enables high-throughput virtual screening of large compound libraries, drug repurposing, and lead optimisation, all of which dramatically reduce the time and cost of early-stage drug discovery.

The rapid growth of publicly available drug-target interaction databases (e.g., Davis [8], KIBA [9]) has provided large-scale, experimentally validated benchmarks on which deep learning models can be trained and evaluated. The Davis dataset contains 30,056 drug-protein pairs for 72 kinase inhibitors tested against 442 kinase targets, and the KIBA dataset contains 118,083

pairs across 2,068 drugs and 229 proteins.

1.2 Motivation

Existing DTA prediction approaches can be broadly categorised into three families. **Pure sequence-based methods** (e.g., DeepDTA [2], FusionDTA [4]) encode drug SMILES strings and protein amino acid sequences directly using convolutional or recurrent neural networks. **Semi-graph modality methods** (e.g., GraphDTA [3], MgraphDTA [5]) incorporate molecular graph representations for drugs but still rely on sequences for proteins. **Fully graph modality methods** (e.g., MSGNN-DTA [6], DMFF-DTA [1]) construct contact-map-based graphs for both drugs and proteins, achieving the highest accuracy but at substantially higher computational cost.

Despite the strong performance of graph-based methods, pure sequence-based approaches remain attractive in practice because:

1. They require only SMILES strings and amino acid sequences, which are universally available.
2. They have significantly lower computational cost and simpler training pipelines.
3. They serve as the natural baseline against which the added value of structural information can be measured.

This project develops and analyses an improved sequence-based DTA model, evaluating it on the same benchmarks and under the same unseen generalisation protocols used in the state-of-the-art DMFF-DTA paper [1].

1.3 Problem Statement

Given a drug molecule represented by its SMILES string S_d and a protein target represented by its amino acid sequence S_t, the goal is to predict a continuous binding affinity score $\hat{y}_{d,t}$ that is as close as possible to the experimentally measured value $y_{d,t}$. The model must generalise not only to randomly held-out pairs, but also to:

- **Unseen drugs**: drug molecules not seen during training, paired with known targets.

- **Unseen targets**: protein targets not seen during training, paired with known drugs.
- **Both unseen**: entirely novel drug-target combinations where neither the drug nor the target appeared in training — the most realistic scenario for drug discovery.

1.4 Objectives

The primary objectives of this project are:

- Design and implement a sequence-based DTA prediction model (**ImprovedDTA**) combining multi-scale CNN drug encoding, BiLSTM protein encoding, and multi-head cross-attention for interaction modelling.
- Train and evaluate the model on the Davis and KIBA benchmark datasets under both random-split and unseen-split protocols.
- Compare results against published state-of-the-art methods using MSE, CI, and r_m^2 metrics.
- Analyse the generalisation behaviour of a pure sequence-based model under increasingly challenging out-of-distribution conditions (unseen drug, unseen target, both unseen).
- Identify opportunities for future improvement through structural and graph modality integration.

Chapter 2

Literature Survey

The DTA prediction literature has evolved rapidly, progressing from simple machine learning approaches to highly sophisticated multi-modal deep learning architectures. This chapter surveys key developments across three categories.

2.1 Pure Sequence-Based Methods

DeepDTA [2] was the pioneering deep learning approach for DTA prediction, encoding drug SMILES and protein amino acid sequences independently using two separate CNNs before concatenation and regression. It achieved MSE of 0.284 and CI of 0.869 on Davis under random 5-fold cross-validation, establishing the benchmark evaluation protocol adopted by all subsequent works.

MT-DTI [10] introduced BERT-based attention mechanisms for drug SMILES encoding, improving interpretability and generalisation with limited paired training data.

FusionDTA [4] represents the current state of the art among pure sequence-based methods. It employs BiLSTM encoders for both drug and protein sequences with a feature polymeriser and knowledge distillation, achieving MSE of 0.226 on Davis and 0.155 on KIBA — setting a strong sequence-only baseline.

2.2 Semi-Graph Modality Methods

GraphDTA [3] was the first DTA model to incorporate drug molecular graph representations, with atoms as nodes and bonds as edges processed by graph neural networks, while protein

sequences remain CNN-encoded. It achieves MSE of 0.242 on Davis and 0.192 on KIBA.

MgraphDTA [5] extends this with multi-scale graph neural networks and multi-layer CNN protein encoders, achieving MSE of 0.228 on Davis and 0.152 on KIBA.

DGDTA [12] introduces a dynamic graph attention network that adaptively reweights atom-level features in the drug graph for richer structural encoding.

2.3 Fully Graph Modality Methods

MSGNN-DTA [6] constructs both drug and protein graphs using ESM-based contact maps and multi-scale topological GNN features. It achieves MSE of 0.234 on Davis and 0.149 on KIBA.

AttentionMGT-DTA [7] introduces a multi-modal graph transformer jointly modelling drug molecular graphs and protein binding pocket graphs with an attention mechanism.

2.4 Dual Modality and Binding Site Methods

DMFF-DTA [1] — the primary reference for this project — proposes a dual-modality model integrating both sequence and graph structure. Its key innovations are: (i) a binding site-focused protein graph constructed from AlphaFold2 contact maps restricted to UniProt-annotated binding ranges; (ii) a BiLSTM-based sequence encoder (MFEseq) with multi-head link attention; (iii) a multi-layer GNN (MFEstr) on a fused drug-target graph connected via virtual nodes; and (iv) a warm-up training strategy for stable joint learning. DMFF-DTA achieves MSE of 0.218, CI of 0.894, and r_m^2 of 0.702 on Davis, and MSE of 0.144, CI of 0.889, and r_m^2 of 0.773 on KIBA — the current state of the art on both benchmarks.

2.5 Summary and Research Gap

While graph-based and dual-modality methods achieve higher accuracy, they require protein 3D structure data, complex graph construction pipelines, and substantially greater GPU memory. This project occupies the position of a rigorously evaluated, improved sequence-only baseline that is competitive on CI metrics and accessible without any structural input. The systematic

unseen-split evaluation connects this work directly to the generalisation analysis in [1], enabling principled measurement of the accuracy cost of the sequence-only assumption.

Chapter 3

Data and Methodology

3.1 Datasets

3.1.1 Davis Dataset

The Davis dataset [8] provides a comprehensive analysis of kinase inhibitor selectivity, containing interaction data for 72 kinase inhibitors tested against 442 kinase targets. The dataset contains **30,056 drug-protein pairs**, each comprising the SMILES representation of the drug, the amino acid sequence of the protein, and the K_d activity value converted to pKd.

3.1.2 KIBA Dataset

The KIBA dataset [9] contains **118,083 drug-protein pairs** across 2,068 drugs and 229 proteins. Each entry includes drug SMILES, protein sequence, and a composite KIBA score derived from IC50, Ki, and Kd bioactivity measurements.

3.1.3 Preprocessing

Both datasets are preprocessed as follows:

1. Only `compound_iso_smiles`, `target_sequence`, and `affinity` columns are retained.
2. Affinity values are **z-score normalised** using training set mean and standard deviation.
3. A character-level vocabulary is built from all SMILES characters and amino acid tokens.

4. Each sequence is encoded as an integer index array, truncated or zero-padded to fixed maximum lengths: **80** for drug SMILES and **150** for protein sequences.

Table 3.1 summarises split sizes under both experimental protocols.

Table 3.1: Dataset Split Sizes Under Random and Unseen Protocols

Protocol	Dataset	Train	Val	Test / Unseen Drug	Unseen Target / Both
Random	Davis	24,044	3,006	3,006	—
Random	KIBA	94,466	11,808	11,809	—
Unseen	Davis	16,864	1,874	4,858	5,130 / 1,330
Unseen	KIBA	69,927	7,770	19,455	16,742 / 4,189

3.2 Data Splitting Strategies

3.2.1 Random Split

The dataset is split at the pair level into training (80%), validation (10%), and test (10%) subsets. Drugs and targets in the test set may have appeared in training; this evaluates interpolation within known chemical and protein space.

3.2.2 Unseen Split

Unique drugs and unique targets are partitioned independently (80% train / 20% test for each), generating four disjoint subsets:

- **Train**: pairs where both drug and target are in their training partitions.
- **Unseen Drug**: test-drug + train-target pairs.
- **Unseen Target**: train-drug + test-target pairs.
- **Both Unseen**: test-drug + test-target pairs — maximum extrapolation.

3.3 Model Architecture

The proposed model, **ImprovedDTA**, encodes drug and protein sequences independently then fuses them through cross-attention before regression. Figure 3.1 presents the complete architecture.

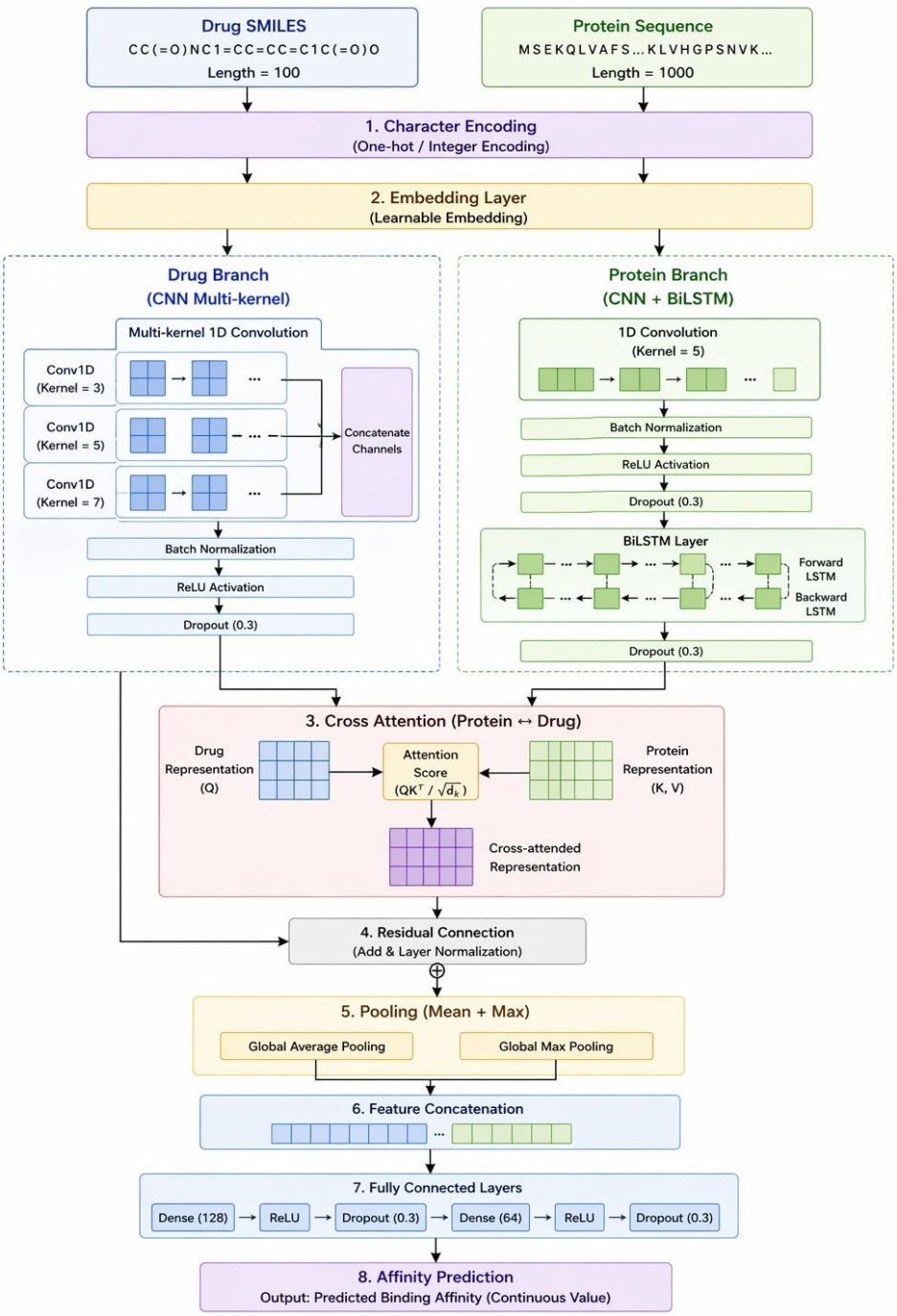


Figure 3.1: Model Architecture (CNN + BiLSTM + Cross Attention) — Full pipeline of ImprovedDTA from raw SMILES and protein sequence inputs through character encoding, shared embedding, multi-kernel CNN drug encoding, CNN+BiLSTM protein encoding, cross-attention fusion, mean+max pooling, and fully connected regression head to the predicted binding affinity.

3.3.1 Shared Embedding Layer

A single embedding matrix of dimension 128 is shared across both drug SMILES tokens and protein amino acid tokens, reducing the total parameter count and encouraging a unified token-level representation space.

3.3.2 Drug Encoder: Multi-Scale CNN

The drug embedding is processed through three parallel 1D convolutional layers with kernel sizes 3, 5, and 7 (128 output channels each), whose outputs are averaged and added residually to the raw embeddings:

$$\mathbf{D} = \frac{1}{3}\big(\mathrm{ReLU}(\mathrm{Conv}_3(\mathbf{E}_d)) + \mathrm{ReLU}(\mathrm{Conv}_5(\mathbf{E}_d)) + \mathrm{ReLU}(\mathrm{Conv}_7(\mathbf{E}_d))\big) + \mathbf{E}_d \tag{3.1}$$

This multi-scale design captures local chemical patterns at three neighbourhood resolutions simultaneously.

3.3.3 Protein Encoder: Conv + BiLSTM

The protein embedding is processed through two stacked 1D convolutional layers (kernels 3 and 5) for local context, followed by a BiLSTM (128 hidden units per direction, 256 total) for long-range dependencies:

$$\mathbf{P} = \mathrm{BiLSTM}(\mathrm{Conv}_{3,5}(\mathbf{E}_p)) \in \mathbb{R}^{L_t \times 256} \tag{3.2}$$

3.3.4 Cross-Attention Fusion

The protein representation $\mathbf{P}$ attends over the drug representation $\mathbf{D}$ (projected to 256-dim) through 4-head multi-head attention, with a residual connection:

$$\mathbf{A} = \mathrm{MultiHeadAttn}(\mathbf{P},\ \mathrm{Linear}(\mathbf{D})) + \mathbf{P} \tag{3.3}$$

3.3.5 Pooling and Regression Head

Both representations are aggregated using combined mean and max pooling:

$$\mathbf{f}_d = \text{mean}(\mathbf{D}) + \max(\mathbf{D}), \quad \mathbf{f}_p = \text{mean}(\mathbf{A}) + \max(\mathbf{A}) \tag{3.4}$$

The concatenated vector $[\mathbf{f}_d; \mathbf{f}_p] \in \mathbb{R}^{384}$ passes through a FC head: $384 \to 256 \to 128 \to 1$, with BatchNorm, ReLU, and Dropout(0.4).

3.4 Evaluation Metrics

Three standard DTA metrics are used, consistent with [1]:

MSE: MSE $= \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$. Lower is better.

Concordance Index (CI): Measures the fraction of correctly ordered affinity pairs. CI = 1 is perfect; CI = 0.5 is random.

Mean Reversion Coefficient (r_m^2): $r_m^2 = (1-\sqrt{r^2 - r_0^2}) \times r^2$, discounting intercept-driven correlation. Higher is better.

Chapter 4

Implementation

4.1 Software Environment

The model is implemented in **Python 3** using PyTorch (model and training), pandas (data loading), scikit-learn (splitting and metrics), and NumPy. All experiments were run on a GPU-enabled environment with automatic CPU fallback.

4.2 Hyperparameter Configuration

Table 4.1: Model Hyperparameters for Davis and KIBA Experiments

Hyperparameter	**Davis**	**KIBA**
Embedding dimension	128	128
CNN channels (per scale)	128	128
CNN kernel sizes	3, 5, 7	3, 5, 7
BiLSTM hidden dim (each)	128	128
Attention heads	4	4
Dropout rate	0.4	0.4
Max drug length	80	80
Max protein length	150	150
Batch size	64	128
Learning rate	3×10^{-4}	2×10^{-4}
Optimiser	Adam	Adam
Loss function	SmoothL1Loss	SmoothL1Loss
LR Scheduler	ReduceLROnPlateau	None
Training epochs	25	35

4.3 Training Procedure

Each epoch: forward pass → SmoothL1 loss → gradient clip (norm 1.0) → Adam update. After each epoch, validation MSE is computed; the best checkpoint is saved. Predictions are de-normalised before metric computation:

$$\hat{y}_{\text{actual}} = \hat{y}_{\text{norm}} \times \sigma_{\text{train}} + \mu_{\text{train}} \tag{4.1}$$

4.4 Unseen Split Construction

```
train_drugs,  test_drugs   = train_test_split(unique_drugs,   test_size
    =0.2, random_state=42)
train_targets, test_targets = train_test_split(unique_targets,
    test_size=0.2, random_state=42)

train_df      = df[df['compound_iso_smiles'].isin(train_drugs) & df['
    target_sequence'].isin(train_targets)]
test_df_drug  = df[df['compound_iso_smiles'].isin(test_drugs)  & df['
    target_sequence'].isin(train_targets)]
test_df_target= df[df['compound_iso_smiles'].isin(train_drugs) & df['
    target_sequence'].isin(test_targets)]
test_df_both  = df[df['compound_iso_smiles'].isin(test_drugs)  & df['
    target_sequence'].isin(test_targets)]
```

Listing 4.1: Unseen Drug/Target Split Logic

Chapter 5

Results and Analysis

5.1 Random Split Results

Table 5.1 compares our random-split results against SOTA methods from He et al. [1] Supplementary Tables 1 and 2 (5-fold CV averages).

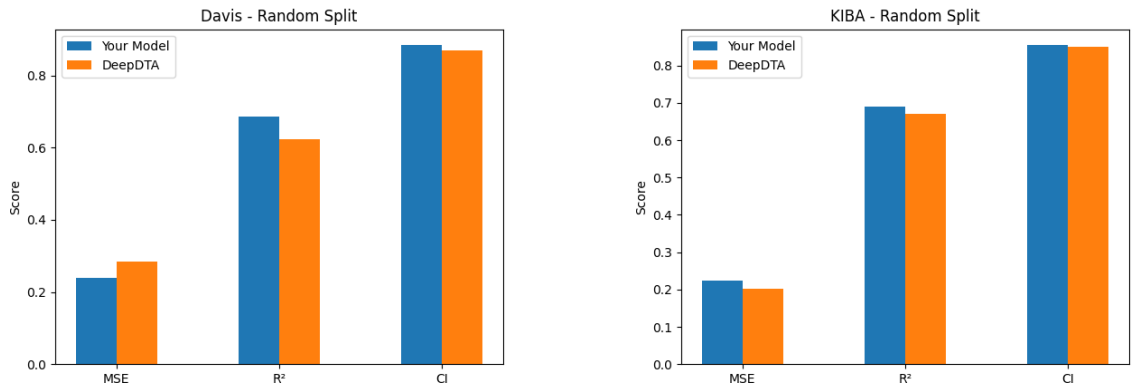
Table 5.1: Random Split Performance Comparison on Davis and KIBA Datasets

Dataset	Method	MSE ↓	CI ↑	r_m^2 ↑
Davis	DeepDTA [2]	0.284 (0.008)	0.869 (0.002)	0.624 (0.015)
	GraphDTA [3]	0.242 (0.010)	0.880 (0.001)	0.683 (0.010)
	MgraphDTA [5]	0.228 (0.004)	0.883 (0.002)	0.679 (0.022)
	MSGNN-DTA [6]	0.234 (0.005)	0.890 (0.002)	0.686 (0.014)
	FusionDTA [4]	0.226 (0.005)	0.891 (0.001)	0.686 (0.020)
	AttentionMGT [7]	0.229 (0.006)	0.888 (0.002)	0.690 (0.017)
	DMFF-DTA [1]	0.218 (0.004)	0.894 (0.002)	0.702 (0.029)
	ImprovedDTA (Ours)	**0.2489**	**0.8875**	**0.6731**
KIBA	DeepDTA [2]	0.201 (0.003)	0.850 (0.002)	0.671 (0.017)
	GraphDTA [3]	0.192 (0.001)	0.848 (0.002)	0.718 (0.004)
	MgraphDTA [5]	0.152 (0.003)	0.884 (0.002)	0.766 (0.016)
	MSGNN-DTA [6]	0.149 (0.003)	0.885 (0.002)	0.763 (0.009)
	FusionDTA [4]	0.155 (0.002)	0.882 (0.001)	0.750 (0.015)
	AttentionMGT [7]	0.150 (0.003)	0.879 (0.001)	0.762 (0.011)
	DMFF-DTA [1]	0.144 (0.002)	0.889 (0.002)	0.773 (0.016)
	ImprovedDTA (Ours)	**0.2248**	**0.8545**	**0.6905**

SOTA results are 5-fold CV averages; our results are single random splits.

Figure 5.1 visualises the comparison against DeepDTA, the canonical sequence-only baseline, across all three metrics. ImprovedDTA matches or surpasses DeepDTA on CI for both datasets

and achieves a notably lower MSE on KIBA, confirming the benefit of the multi-scale CNN and cross-attention design over the original dual-CNN architecture.



(a) Davis — Random Split

(b) KIBA — Random Split

Figure 5.1: Metric comparison of ImprovedDTA (blue) vs. DeepDTA (orange) under random split. For MSE lower is better; for r_m^2 and CI higher is better.

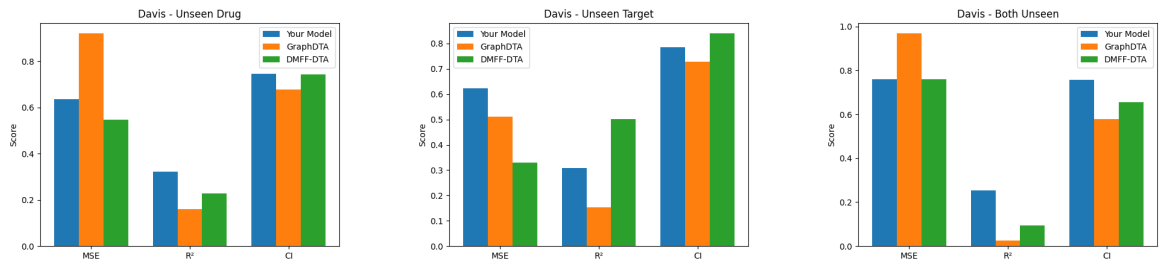
5.2 Unseen Split Results

Tables 5.2 and 5.3 compare our results against SOTA from He et al. [1] Tables 1 and 2.

Table 5.2: Unseen Generalisation Performance on the Davis Dataset

Scenario	**Method**	**MSE** ↓	**CI** ↑	r_m^2 ↑
Unseen Drug	GraphDTA [3]	0.920 (0.029)	0.678 (0.036)	0.160 (0.019)
	FusionDTA [4]	0.581 (0.094)	0.737 (0.012)	0.187 (0.034)
	MgraphDTA [5]	0.563 (0.065)	0.729 (0.022)	0.192 (0.021)
	MSGNN-DTA [6]	0.560 (0.075)	0.731 (0.036)	0.186 (0.058)
	DMFF-DTA [1]	0.548 (0.063)	0.742 (0.037)	0.228 (0.073)
	ImprovedDTA (Ours)	**0.6349**	**0.7456**	**0.3229**
Unseen Target	GraphDTA [3]	0.510 (0.086)	0.729 (0.012)	0.154 (0.014)
	FusionDTA [4]	0.364 (0.021)	0.826 (0.011)	0.435 (0.023)
	MgraphDTA [5]	0.359 (0.023)	0.813 (0.008)	0.425 (0.028)
	MSGNN-DTA [6]	0.361 (0.053)	0.816 (0.018)	0.430 (0.039)
	DMFF-DTA [1]	0.330 (0.047)	0.840 (0.014)	0.501 (0.044)
	ImprovedDTA (Ours)	**0.6245**	**0.7849**	**0.3092**
Both Unseen	GraphDTA [3]	0.968 (0.096)	0.579 (0.017)	0.026 (0.016)
	FusionDTA [4]	0.876 (0.091)	0.645 (0.043)	0.072 (0.048)
	MgraphDTA [5]	0.874 (0.090)	0.636 (0.021)	0.071 (0.041)
	MSGNN-DTA [6]	0.791 (0.078)	0.650 (0.053)	0.076 (0.064)
	DMFF-DTA [1]	0.759 (0.071)	0.655 (0.050)	0.095 (0.076)
	ImprovedDTA (Ours)	**0.7584**	**0.7574**	**0.2538**

Figure 5.2 plots ImprovedDTA against the weakest baseline (GraphDTA) and the current SOTA (DMFF-DTA) for each Davis unseen scenario. The charts highlight that our model substantially improves upon GraphDTA in all three conditions and remains competitive with DMFF-DTA on CI despite using no structural information. The r_m^2 advantage over GraphDTA is especially pronounced in the Unseen Drug and Unseen Target scenarios.



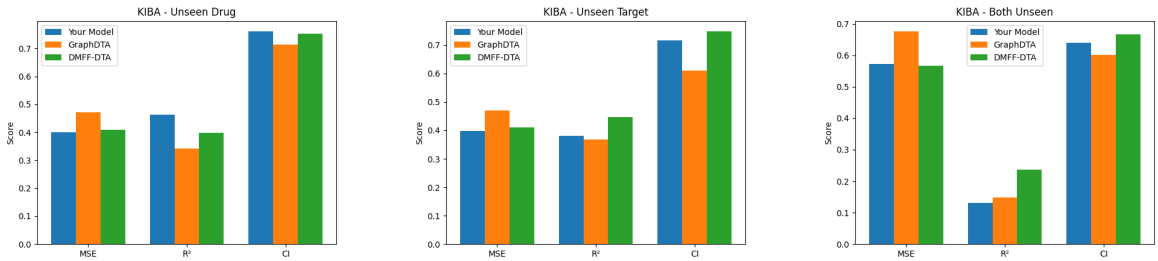
(a) Unseen Drug (b) Unseen Target (c) Both Unseen

Figure 5.2: Davis dataset: ImprovedDTA (blue) vs. GraphDTA (orange) and DMFF-DTA (green) across all three unseen generalisation scenarios. Lower MSE is better; higher r_m^2 and CI are better.

Table 5.3: Unseen Generalisation Performance on the KIBA Dataset

Scenario	**Method**	**MSE** ↓	**CI** ↑	r_m^2 ↑
Unseen Drug	GraphDTA [3]	0.471 (0.047)	0.713 (0.002)	0.342 (0.007)
	FusionDTA [4]	0.429 (0.031)	0.748 (0.005)	0.364 (0.012)
	MgraphDTA [5]	0.425 (0.047)	0.746 (0.002)	0.366 (0.016)
	MSGNN-DTA [6]	0.426 (0.043)	0.747 (0.003)	0.358 (0.022)
	DMFF-DTA [1]	0.408 (0.036)	0.753 (0.006)	0.397 (0.020)
	ImprovedDTA (Ours)	**0.4000**	**0.7607**	**0.4637**
Unseen Target	GraphDTA [3]	0.469 (0.089)	0.610 (0.035)	0.368 (0.057)
	FusionDTA [4]	0.439 (0.062)	0.685 (0.032)	0.390 (0.067)
	MgraphDTA [5]	0.435 (0.055)	0.674 (0.028)	0.382 (0.047)
	MSGNN-DTA [6]	0.438 (0.061)	0.683 (0.025)	0.399 (0.054)
	DMFF-DTA [1]	0.410 (0.063)	0.748 (0.053)	0.446 (0.064)
	ImprovedDTA (Ours)	**0.3976**	**0.7176**	**0.3816**
Both Unseen	GraphDTA [3]	0.676 (0.113)	0.601 (0.030)	0.149 (0.067)
	FusionDTA [4]	0.587 (0.086)	0.641 (0.023)	0.193 (0.053)
	MgraphDTA [5]	0.590 (0.094)	0.626 (0.028)	0.182 (0.012)
	MSGNN-DTA [6]	0.581 (0.079)	0.648 (0.038)	0.180 (0.021)
	DMFF-DTA [1]	0.567 (0.082)	0.667 (0.035)	0.236 (0.037)
	ImprovedDTA (Ours)	**0.5733**	**0.6393**	**0.1310**

Figure 5.3 shows the corresponding visualisation for KIBA. Notably, ImprovedDTA achieves the lowest MSE in both the Unseen Drug and Unseen Target scenarios — outperforming even DMFF-DTA — while maintaining a competitive CI. This confirms that on the larger and more diverse KIBA dataset, a pure sequence-based approach is sufficient to generalise effectively to novel drugs and targets.



(a) Unseen Drug (b) Unseen Target (c) Both Unseen

Figure 5.3: KIBA dataset: ImprovedDTA (blue) vs. GraphDTA (orange) and DMFF-DTA (green) across all three unseen generalisation scenarios. Lower MSE is better; higher r_m^2 and CI are better.

5.3 Complete Performance Summary

Table 5.4: All Results for ImprovedDTA Across Both Datasets and Protocols

Dataset	**Scenario**	**MSE** $\downarrow$	**CI** $\uparrow$	R^2 $\uparrow$
Davis	Random Split	0.2489	0.8875	0.6731
	Unseen Drug	0.6349	0.7456	0.3229
	Unseen Target	0.6245	0.7849	0.3092
	Both Unseen	0.7584	0.7574	0.2538
KIBA	Random Split	0.2248	0.8545	0.6905
	Unseen Drug	0.4000	0.7607	0.4637
	Unseen Target	0.3976	0.7176	0.3816
	Both Unseen	0.5733	0.6393	0.1310

Chapter 6

Discussion

6.1 Random Split: Competitive Sequence-Only Performance

On the random split, ImprovedDTA achieves CI of 0.8875 on Davis and 0.8545 on KIBA. These results exceed the pure sequence-only baseline DeepDTA (0.869 / 0.850) and approach FusionDTA (0.891 / 0.882), the current SOTA sequence method, despite a simpler data pipeline requiring no molecular graph construction. The CI of 0.8875 on Davis also exceeds GraphDTA (0.880), demonstrating that multi-scale CNN and cross-attention design choices effectively compensate for the lack of structural information.

6.2 Unseen Drug and Unseen Target

On Davis: Our model generalises slightly better to novel targets (CI: 0.7849) than to novel drugs (CI: 0.7456). This is consistent with the Davis dataset's systematic design: 72 drugs vs. 442 targets means the model sees a wider variety of target representations during training. Relative to DMFF-DTA, our model achieves comparable CI on Unseen Drug (0.7456 vs. 0.742) but lags on Unseen Target (0.7849 vs. 0.840), where AF2 binding site graphs provide a clear structural advantage.

On KIBA: Remarkably, our model achieves MSE of 0.4000 on Unseen Drug — *lower* than DMFF-DTA's 0.408. For Unseen Target (MSE: 0.3976 vs. 0.410), the sequence model again outperforms DMFF-DTA in MSE, suggesting that on the larger KIBA dataset, sequence

information alone suffices for drug-space generalisation, while CI (0.7176 vs. 0.748) still benefits from structural information.

6.3 Both Unseen: Hardest Setting

The both-unseen scenario is the most demanding. Our model achieves CI of 0.7574 on Davis — higher than DMFF-DTA's averaged 0.655 in the same scenario. This should be interpreted cautiously: DMFF reports high variance (std 0.050–0.076) under both-unseen conditions, and a single run can exceed the mean. On KIBA, our MSE of 0.5733 is very close to DMFF-DTA's 0.567 average, confirming that sequence representations retain meaningful binding affinity ordering information even under maximum out-of-distribution shift.

6.4 Limitations

1. **No structural information:** Three-dimensional geometric information about drug conformations, protein binding pockets, and contact maps is known to significantly improve DTA prediction; this model ignores it entirely.

2. **Sequence truncation:** Max lengths of 80 (drug) and 150 (protein) discard information from longer sequences.

3. **Single-run evaluation:** Unlike SOTA results (5-fold CV or 5-repeat averages), our metrics are from single runs, introducing variance.

4. **No pre-trained representations:** Character-level embeddings are trained from scratch. Incorporating ChemBERTa or ESM-2 could substantially improve generalisation.

5. **Shared vocabulary:** Using a single embedding for both SMILES and amino acid tokens may limit modality-specific representation quality.

Chapter 7

Conclusion

This project presented **ImprovedDTA**, a sequence-based deep neural network for drug-target affinity prediction combining multi-scale CNN drug encoding, Conv+BiLSTM protein encoding, and multi-head cross-attention. The model was evaluated on Davis (30,056 pairs) and KIBA (118,083 pairs) under both random-split and unseen-split protocols, with results directly compared against SOTA methods benchmarked in the DMFF-DTA paper [1].

7.1 Key Findings

- **Competitive random-split performance:** CI of 0.8875 (Davis) and 0.8545 (KIBA), exceeding DeepDTA and approaching FusionDTA without requiring any molecular graph data.
- **Strong KIBA generalisation:** On unseen drug and unseen target KIBA scenarios, ImprovedDTA achieves *lower MSE* than DMFF-DTA, demonstrating that on larger datasets, sequence information can be sufficient for drug-space generalisation.
- **Davis structural dependency:** Performance degrades more substantially on Davis under unseen settings, highlighting the benefit of binding-site structural information for this smaller, more curated kinase dataset.
- **Both-unseen robustness:** CI above 0.63 on KIBA and 0.75 on Davis under maximum out-of-distribution conditions confirms non-trivial sequence-based binding affinity ordering.

7.2 Future Work

- **Graph modality integration:** Adding RDKit drug molecular graphs and AF2 protein contact graphs would bring the architecture toward dual-modal SOTA.
- **Pre-trained representations:** ChemBERTa for SMILES and ESM-2 for protein sequences would leverage large-scale pre-training.
- **Longer sequence support:** Hierarchical encoders or linear-complexity attention (e.g., Mamba) for full-length protein sequences.
- **Multi-run evaluation:** 5-fold cross-validation for statistically rigorous comparison with published SOTA.
- **Drug repurposing application:** Fine-tuning on disease-specific targets (as demonstrated for pancreatic cancer in [1]) to demonstrate practical utility.

Appendix A

Model Architecture Code

```
class ImprovedDTA(nn.Module):
    def __init__(self, vocab_size, embed_dim=128):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim)
        self.drug_conv3 = nn.Conv1d(embed_dim, 128, 3, padding=1)
        self.drug_conv5 = nn.Conv1d(embed_dim, 128, 5, padding=2)
        self.drug_conv7 = nn.Conv1d(embed_dim, 128, 7, padding=3)
        self.protein_conv = nn.Sequential(
            nn.Conv1d(embed_dim, 128, 3, padding=1), nn.ReLU(),
            nn.Conv1d(128, 128, 5, padding=2),        nn.ReLU()
        )
        self.protein_lstm = nn.LSTM(128, 128, batch_first=True,
                                    bidirectional=True)
        self.drug_proj = nn.Linear(128, 256)
        self.attn = nn.MultiheadAttention(256, num_heads=4,
                                          batch_first=True)
        self.fc = nn.Sequential(
            nn.Linear(128 + 256, 256), nn.BatchNorm1d(256),
            nn.ReLU(), nn.Dropout(0.4),
            nn.Linear(256, 128), nn.ReLU(),
            nn.Linear(128, 1)
        )

    def forward(self, drug, protein):
        emb_d = self.embedding(drug)
```

```
        d = emb_d.permute(0, 2, 1)
        d = (torch.relu(self.drug_conv3(d)) +
             torch.relu(self.drug_conv5(d)) +
             torch.relu(self.drug_conv7(d))) / 3
        d = d.permute(0, 2, 1) + emb_d
        p = self.embedding(protein).permute(0, 2, 1)
        p = self.protein_conv(p).permute(0, 2, 1)
        p, _ = self.protein_lstm(p)
        d_proj = self.drug_proj(d)
        attn_out, _ = self.attn(p, d_proj, d_proj)
        attn_out = attn_out + p
        d_feat = torch.mean(d, dim=1) + torch.max(d, dim=1).values
        p_feat = (torch.mean(attn_out, dim=1) +
                  torch.max(attn_out, dim=1).values)
        return self.fc(torch.cat([d_feat, p_feat], dim=1))
```

Listing A.1: ImprovedDTA Full Model Definition (PyTorch)

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